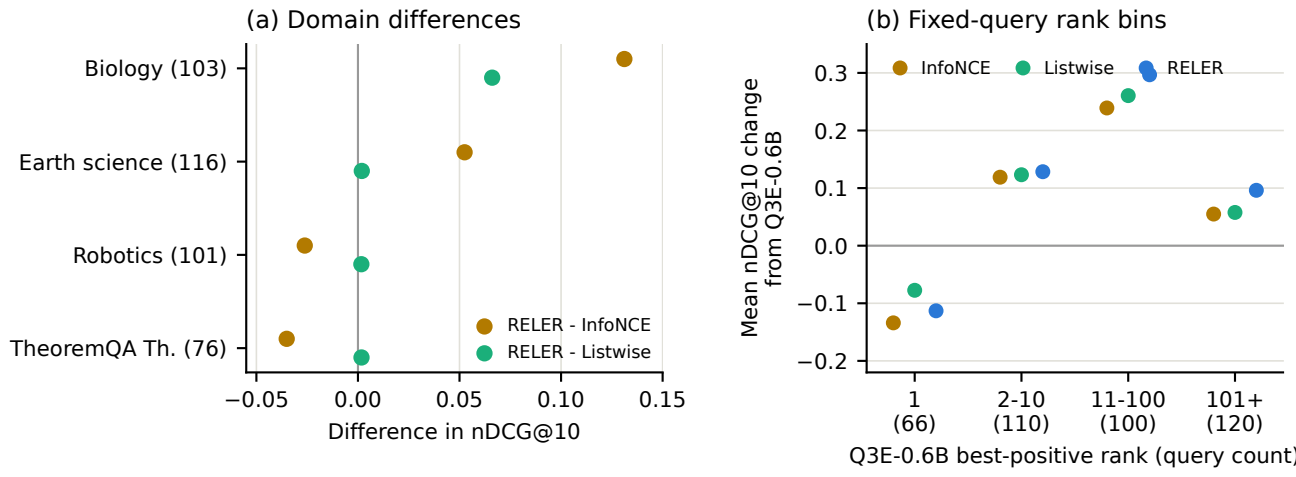
(a) Domain differences
Biology (103)
Earth science (116)
Robotics (101)
TheoremQA Th. (76)
RELER - InfoNCE
RELER - Listwise
−0.05
0.00
0.05
0.10
0.15
Difference in nDCG@10
(b) Fixed-query rank bins
InfoNCE
Listwise
RELER
Mean nDCG@10 change from Q3E-0.6B
0.3
0.2
0.1
0.0
−0.1
−0.2
1 (66)
2-10 (110)
11-100 (100)
101+ (120)
Q3E-0.6B best-positive rank (query count)